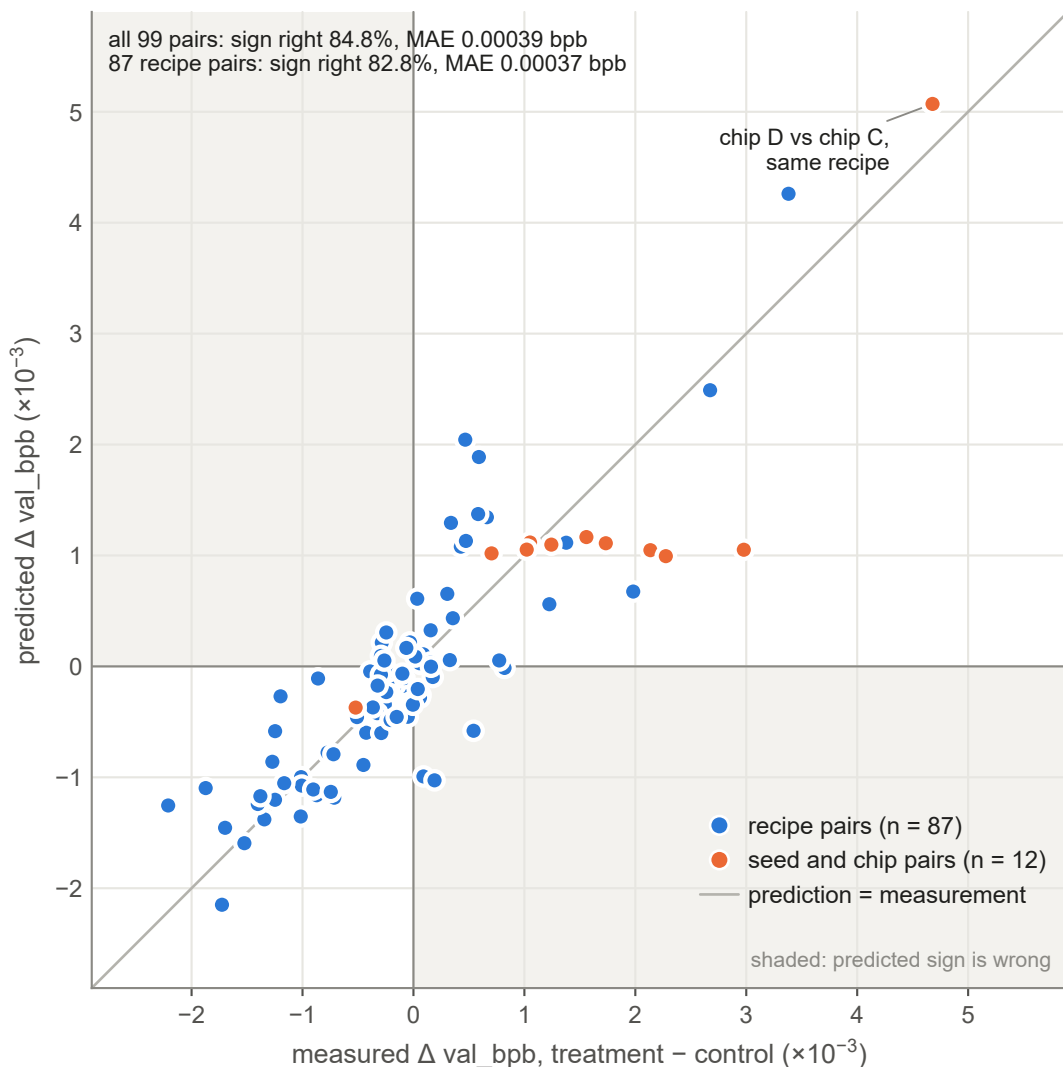


The simulator's quality model on pairs it did not see

Each point is one same-seed chip treatment/control pair (all $n = 99$). The model is refitted without the pair's two runs, then predicts their `val_bpb` difference.



Source: `research/sim-data/runs.jsonl` and `validation-pairs.json`; `ffsim.quality.QualityModel.pair_validation(leave_pair_out=True)`.
97% of measured differences lie within ± 2 predicted sd. Seed and chip pairs measure run-to-run noise, not a recipe effect.
Per-subset numbers: `research/sim-data/quality-validation.md` section 3.